

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	A. Lakshmi
Register Number	192373066

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I A. Lakshmi(192373066) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *lakshmi*

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Aim:

Aim:

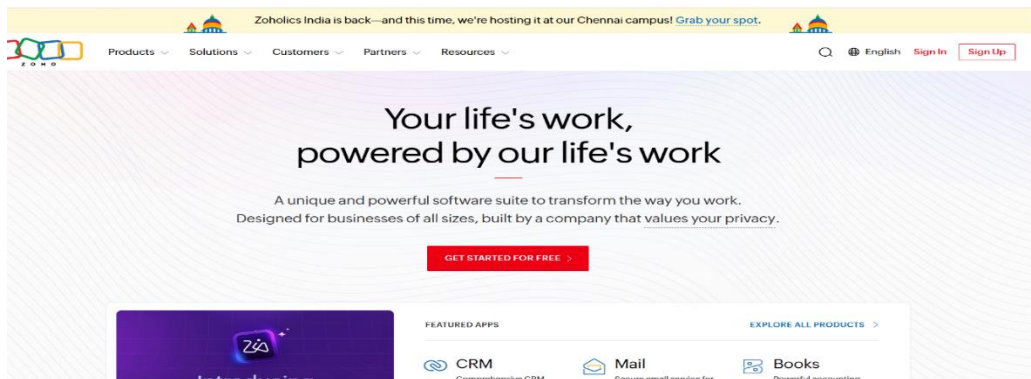
To create a cloud-based Library Book Reservation System using Zoho Creator.

Steps:

1. Create a new application.
2. Create a Book Reservation form.
3. Add fields: Book Title, Author, Publication, Edition, Rack Number, Number of Copies, and Status.
4. Create a report to display available and reserved books.
5. Test the application with sample book records.

6. Implemented

Step 1: Go to zoho.com



STEP 2: Click Create New Application and select Create from Scratch

Create New Application

Create using Zia

Build a functional application effortlessly using GenAI

Create

Create from scratch

Create a new application for your business needs

Create

Create from gallery

Pick a pre-built application that suits your requirement

Pick

Import from file

Import your existing data to create an application

Import

STEP 3: Enter Library Book Reservation System as the application name and click Create Application.

Create from scratch

Application Name *

Library Book Reservation System|

☐ Enable Environment [Learn More](#)

Create Application

Step 4:Add fields: Book Title, Author, Publication, Edition, Rack Number, Number of Copies, and Status

Library Management System

Library Management Syste...

- [Dashboard](#)
- [Issue Records](#)
- + Issue Records**
- [Issue Records Rep...](#)
- [Board](#)
- [Borrowing History](#)
- [Books](#)
- [Students](#)
- [Return Records](#)
- [Dashboard Metrics](#)
- [Items](#)

Issue Records

Student

Book

-Select-

Issue Date

dd-MMM-yyyy

Return Date

dd-MMM-yyyy

Due Date

dd-MMM-yyyy

Status

-Select- v

Fine Amount

#####.##

Returned By

Submit

Reset

Library Management System

Library Management Syste...

Borrowing History

Dashboard

Issue Records

Borrowing History

+ Borrowing History

Borrowing History ...

Board

Books

Students

Return Records

Dashboard Metrics

Items

Student

Book

Issue Date

Return Date

Status

Fine Collected

Submit

Reset

Library Management System

Library Management Syste...

Books

Dashboard

Issue Records

Borrowing History

Books

+ Books

Books Report

Board

Students

Return Records

Dashboard Metrics

Items

Title

Author

Publication

Edition

Year

Rack Number

Copies

Available Copies

Status

Item

Library Management System

Library Management Syste...

Students

Dashboard

Issue Records

Borrowing History

Books

Students

+ Students

Students Report

Board

Return Records

Dashboard Metrics

Items

First Name

Last Name

Email

Phone

Enrollment Date

Status

User

Submit

Reset

ankalalaksh...

https://creatorapp.zoho.in/ankalalakshmi3066.sse_sav/library-management-system#Form:students

STEP 5: Save the form, enter sample booking records, and view them in the Library Book Reservation System.

Library Management System

Trial expires in 2 days Upgrade Edit this application Help

Issue Records Report

Issue ID	Student	Book	Issue Date	Return Date	Due Date	Status	Fine Amount	Returned By
5	2	1 2 3	19-Jul-2026	30-Jul-2026	22-Jul-2026	Returned	\$ 25.75	Emma Brown
4	1 3	2 3	18-Jul-2026	28-Jul-2026	21-Jul-2026	Returned	\$ 0.00	David Lee
3	1 3 4 5	1 2 3 4 5	17-Jul-2026	25-Jul-2026	20-Jul-2026	Issued	\$ 12.00	Carol White
2	1 2 3 5	1 2 3 4	16-Jul-2026	22-Jul-2026	19-Jul-2026	Returned	\$ 5.50	Bob Smith
1	1 2	1	15-Jul-2026	20-Jul-2026	18-Jul-2026	Issued	\$ 0.00	Alice Johnson

Showing 5 of 5

Show Summary

Library Management System

Trial expires in 2 days Upgrade Edit this application Help

Borrowing History Report

Student	Book	Issue Date	Return Date	Status	Fine Collected
1 3	1 2 3 4 5	19-Jul-2026	30-Jul-2026	Returned	\$ 0.00
1 2 4	1 2 3 4	18-Jul-2026	19-Jul-2026	Returned	\$ 12.75
1 2 3 4	1 2 3 5	17-Jul-2026	25-Jul-2026	Completed	\$ 0.00
1 2 3 4	1 2 3	16-Jul-2026	20-Jul-2026	Completed	\$ 5.50
1 3	1 2 3 4	15-Jul-2026	22-Jul-2026	Completed	\$ 0.00

Showing 5 of 5

Show Summary

Library Management System

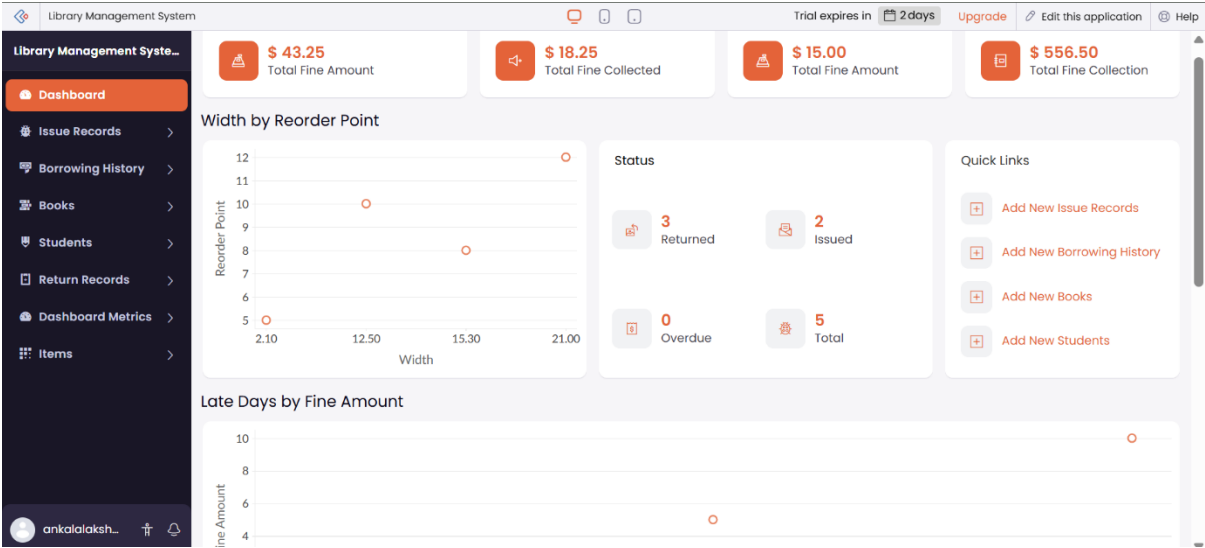
Trial expires in 2 days Upgrade Edit this application Help

Books Report

Book ID	Title	Author	Publication	Edit...	Year	Rack Number	Copies	Available Copies	Status
5	The Catcher in the Rye	J.D. Salinger	Little, Brown		1951	E-520	35	30	Under Maint
4	Pride and Prejudice	Jane Austen	T. Nelson		1813	D-418	25	22	Issued
3	1984	George Orwell	Secker & Warburg		1949	C-312	40	35	Issued
2	To Kill a Mockingbird	Harper Lee	HarperCollins		1960	B-205	30	28	Issued
1	The Great Gatsby	F. Scott Fitzgerald	Scribner		1925	A-101	50	45	Available

Showing 5 of 5

Step 6: Final Dash Board



Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Aim:

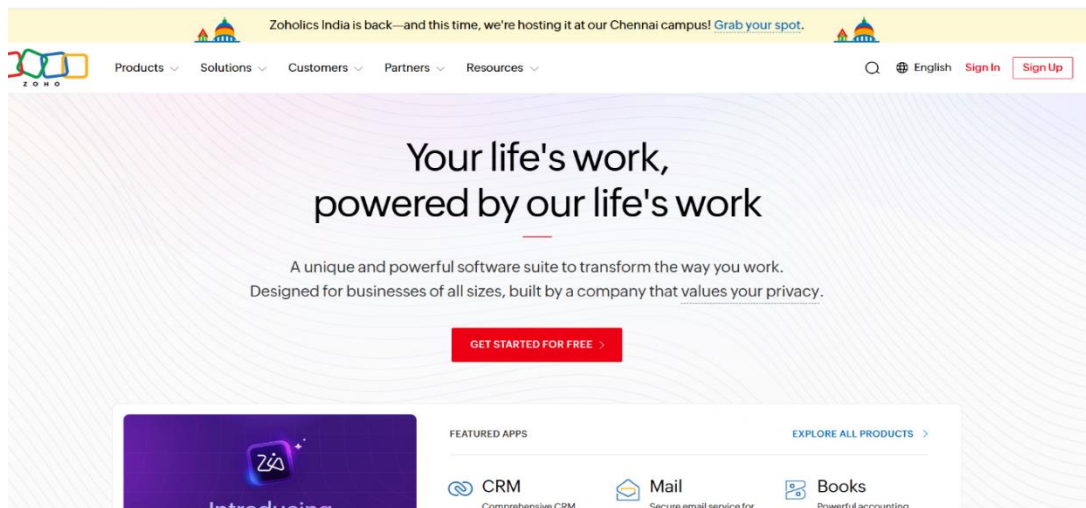
To create a cloud-based Payroll Processing System using Zoho Creator.

Steps:

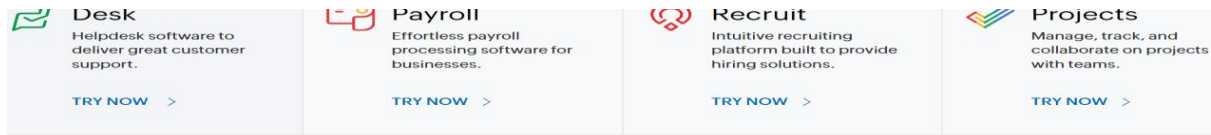
1. Create a new application.
2. Create an Employee Payroll form.
3. Add fields: Employee Name, Experience, Basic Pay, HRA, DA, TA, Gross Pay, and Net Pay.
4. Use formulas to calculate Gross Pay and Net Pay.
5. Save employee details and verify the payroll report.

Implemented:

Step 1: Go to zoho.com



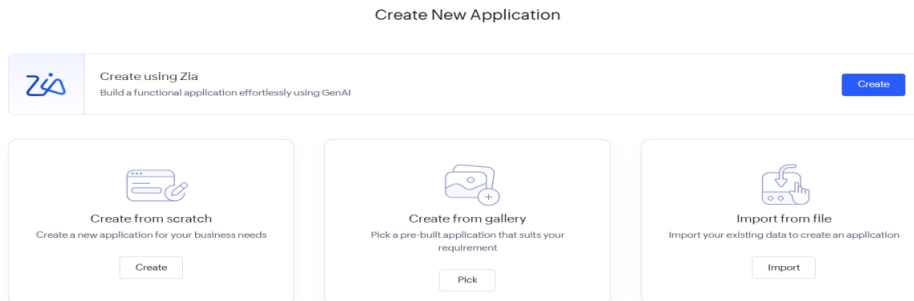
STEP 2: Login to your Zoho account and open Zoho Creator



Apps you've accessed



STEP 3: Click Create New Application and select Create from Scratch



STEP 4: Enter Payroll Processing System as the application name and click Create Application.

Create from scratch

Application Name *

Payroll Processing System

☐ Enable Environment [Learn More](#)

Create Application

Step 5: Add fields: Student Name, Register Number, Subject Marks, Total, Average, Percentage, and Rank.

Payroll Management System

Trial expires in

Payroll Management Syste...

Dashboard

Employees

+ Employees

Employees Report

Board

Attendance Recor...

Employee Salary

Department Sum...

Payroll Calculatio...

Salary Slips

Payroll History

Employees

Employee ID

-Select-

First Name

Last Name

Date of Joining

dd-MMM-yyyy

Date of Birth

dd-MMM-yyyy

Gender

-Select-

Employment Type

-Select-

Department

-Select-

Designation

-Select-

Email Address

Mobile No.

Payroll Management System

Trial expires in

Payroll Management Syste...

Dashboard

Employees

Attendance Recor...

Employee Salary

+ Employee Salary

Employee Salary R...

Department Sum...

Payroll Calculatio...

Salary Slips

Payroll History

Salary Componen...

Employee Salary

Employee ID

-Select-

Component ID

-Select-

Amount

#####.##

Percentage

☐ Is Active

Effective Date

dd-MMM-yyyy

Submit

Reset

Payroll Management System

Trial expires in

Payroll Management Syste...

Dashboard

Employees

Attendance Recor...

Employee Salary

Department Sum...

+ artment Summary

Department Summ...

Payroll Calculatio...

Salary Slips

Payroll History

Salary Componen...

Department Summary

Department ID

-Select-

Period

Total Employees

#####

Total Salary Expense

#####.##

Average Salary

#####.##

Payroll Count

#####

Last Updated

dd-MMM-yyyy HH:mm:ss

Submit

Reset

Payroll Management System

Payroll Calculations

Dashboard

Employees >

Attendance Recor... >

Employee Salary >

Department Sum... >

Payroll Calculatio... >

+ Payroll Calculations

Payroll Calculation...

Board

Salary Slips >

Payroll History >

Employee ID

-Select-

Period

Gross Pay

#####.##

Deductions

#####.##

Net Pay

#####.##

Pay Date

dd-MMM-yyyy

Status

-Select-

Generated By

Generated At

dd-MMM-yyyy HH:mm:ss

Salary Slip

https://

Payroll Management System

Payroll History

Dashboard

Employees >

Attendance Recor... >

Employee Salary >

Department Sum... >

Payroll Calculatio... >

Salary Slips >

Payroll History >

+ Payroll History

Payroll History Rep...

Board

Employee ID

-Select-

Payroll ID

-Select-

Period

Gross Pay

#####.##

Net Pay

#####.##

Pay Date

dd-MMM-yyyy

Status

-Select-

Processed By

Processed At

dd-MMM-yyyy HH:mm:ss

Submit

Reset

STEP 6: Save the form Payroll Processing, enter sample booking records, and view them in the System.

Payroll Management System

Employee Reports Report

Payroll Calculatio... >

Salary Slips >

Payroll History >

Salary Componen... >

Employee Reports >

+ Employee Reports

Employee Reports Report

Salary Reports >

Zoho People Empl... >

Zoho People Depa... >

Zoho People Desig... >

Employee ID

Report Period

Total Present Days

Total Absent Days

Total Leave Days

Attendance Percentage

Last Up

carol.davis@company.com	Jan-Jun-2026	215	0	10	95.65	30-Jul-
david.miller@company.com	Jul-2026	25	3	0	89.65	30-Jul-
david.miller@company.com	Jan-Jun-2026	205	10	15	93.10	30-Jul-
david.miller@company.com	Jul-2026	28	2	1	99.28	30-Jul-
emma.wilson@company.com	Jan-Jun-2026	210	5	10	95.23	30-Jul-

Payroll Management System

Trial expires in 2 days Upgrade Edit this application Help

Payroll Management System...

Payroll Calculation...

Salary Slips

Payroll History

Salary Componen...

Employee Reports

Salary Reports

Salary Reports

Salary Reports Report

Zoho People Empl...

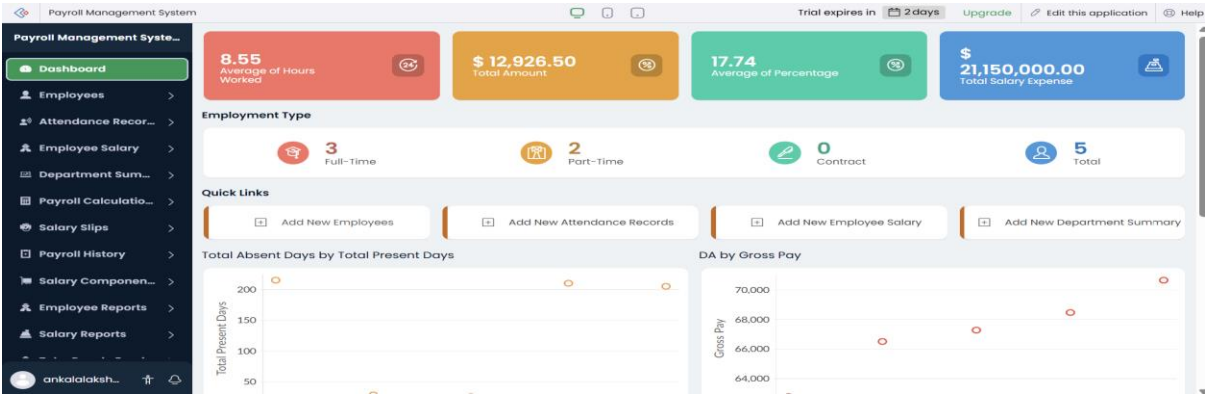
Zoho People Depa...

Zoho People Desig...

Salary Reports Report

Component ID	Period	Total Amount	Average Amount	Max Amount	Min Amount	Employee Count	Last Updated
1	March-2026	\$ 105,000.00	\$ 4,375.00	\$ 7,400.00	\$ 2,700.00	24	30-Jul-2026 14:30:00
2							
3							
1	April-2026	\$ 109,000.00	\$ 4,541.67	\$ 7,600.00	\$ 2,800.00	24	30-Jul-2026 14:15:00
3							
1	May-2026	\$ 112,500.25	\$ 4,687.51	\$ 7,900.00	\$ 2,950.00	24	30-Jul-2026 14:20:00
2							
3							
1	June-2026	\$ 118,000.75	\$ 4,916.70	\$ 8,200.00	\$ 3,100.00	24	30-Jul-2026 14:25:00
3							
1	July-2026	\$ 125,000.50	\$ 5,208.35	\$ 8,500.00	\$ 3,200.00	24	30-Jul-2026 14:30:00
3							

Step 7:Final Dashboard



Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

AIM:

To create a virtual machine with 1 cpu, 2gb ram and 15gb storage disk using a type 2 virtualization software.

PROCEDURE:

STEP 1:Download VMware workstation and installed as type 2hypervisor.

STEP2:Download ubuntu or tiny OS as iso image file.

STEP 3: In VMware workstation->create new VM.

STEP 4: Do the basic configuration settings.

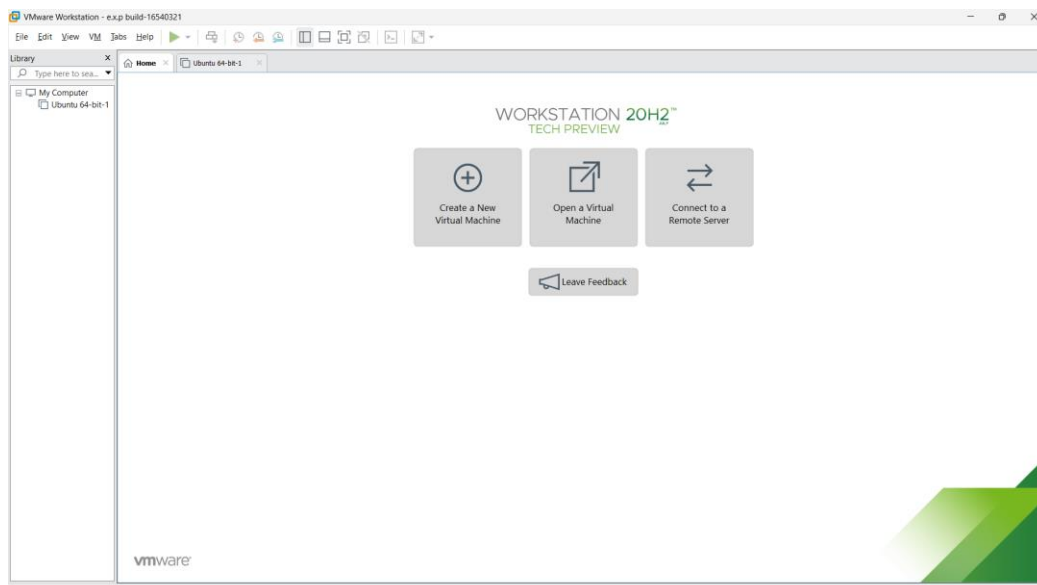
STEP 5: Created tiny OS virtual machine.

STEP 6: Launch the VM.

IMPLEMENTATION:

STEP 1:DOWNLOAD VMWARE WORKSTATIONAND INSTALLED AS

STEP 1:DOWNLOAD VMWARE WORKSTATIONAND INSTALLED AS TYPE 2HYPERVISOR

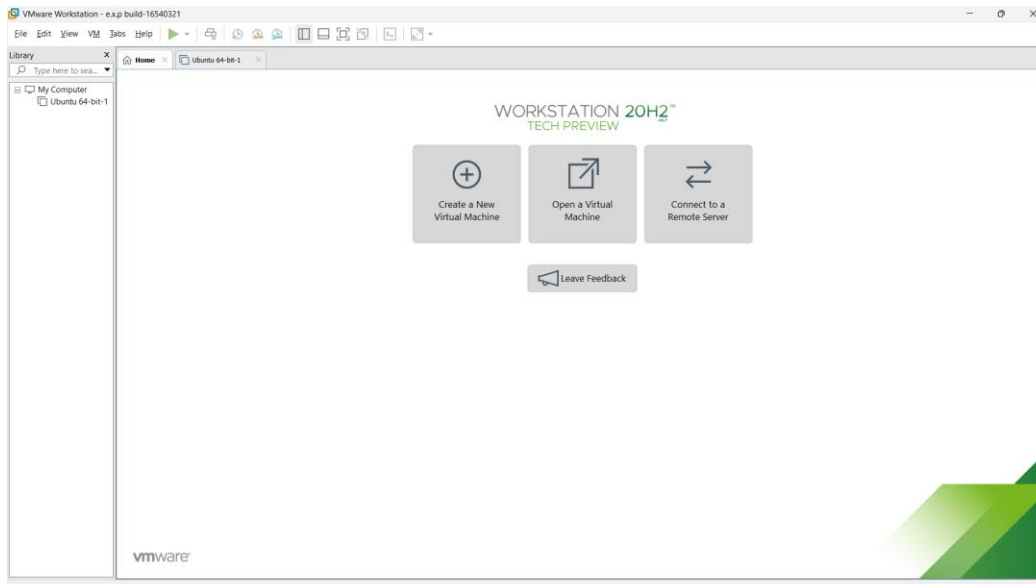


STEP2: DOWLOAD UBUNTU OR TINY OS AS ISO IMAGE FILE

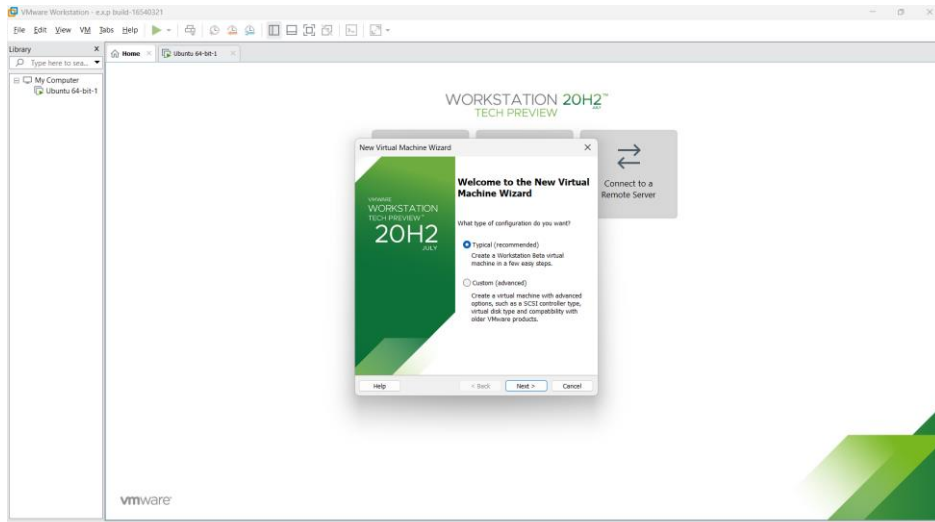
Index of /11.x/x86/release/

../	09-Feb-2020 11:50	-
distribution_files/	03-Dec-2019 11:14	-
src/	01-Apr-2020 07:49	14757888
Core-11.1.iso	01-Apr-2020 07:49	48
Core-11.1.iso.md5.txt	01-Apr-2020 07:49	50639
Core-11.1.iso.zsync	01-Apr-2020 07:49	14757888
Core-current.iso	01-Apr-2020 07:50	216006656
CorePlus-11.1.iso	01-Apr-2020 07:50	52
CorePlus-11.1.iso.md5.txt	01-Apr-2020 07:50	369358
CorePlus-11.1.iso.zsync	01-Apr-2020 07:50	216006656
CorePlus-current.iso	01-Apr-2020 07:50	19922944
TinyCore-11.1.iso	01-Apr-2020 07:50	52
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	68301
TinyCore-11.1.iso.zsync	01-Apr-2020 07:50	19922944
TinyCore-current.iso		

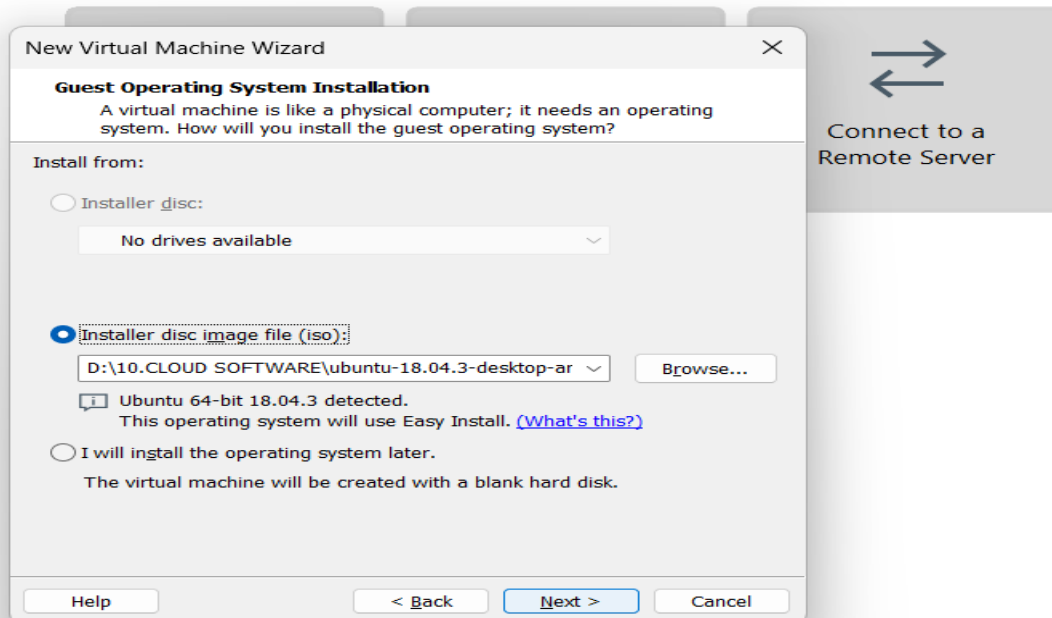
STEP 3: IN VMWARE WORKSTATION->CREATE NEW VM

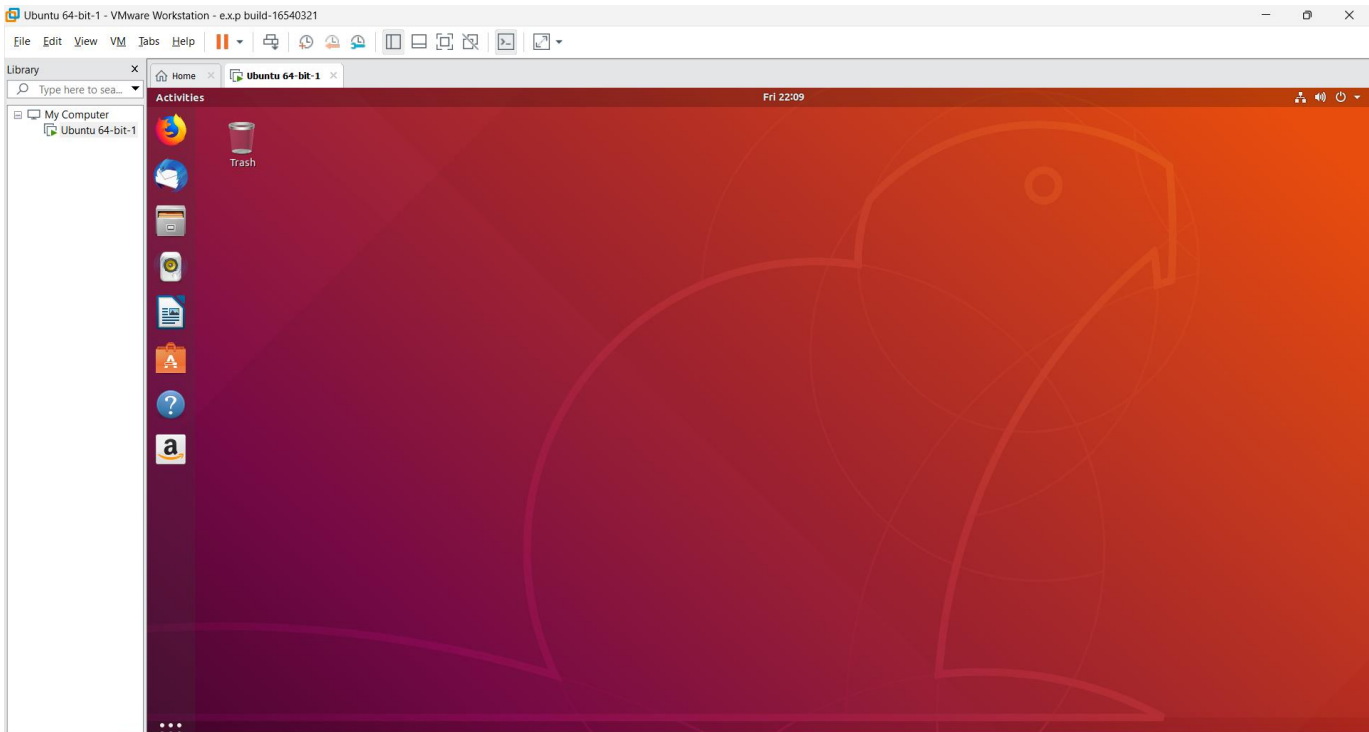


STEP 4: DO THE BASIC CONFIGURATION SETTINGS.



WORKSTATION 20H2TM TECH PREVIEW





Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

AIM:

To create a snapshot of a vm and test it by loading the previous version/cloned vm

PROCEDURE:

STEP 1: GOTO VMWARE WORKSTATION.

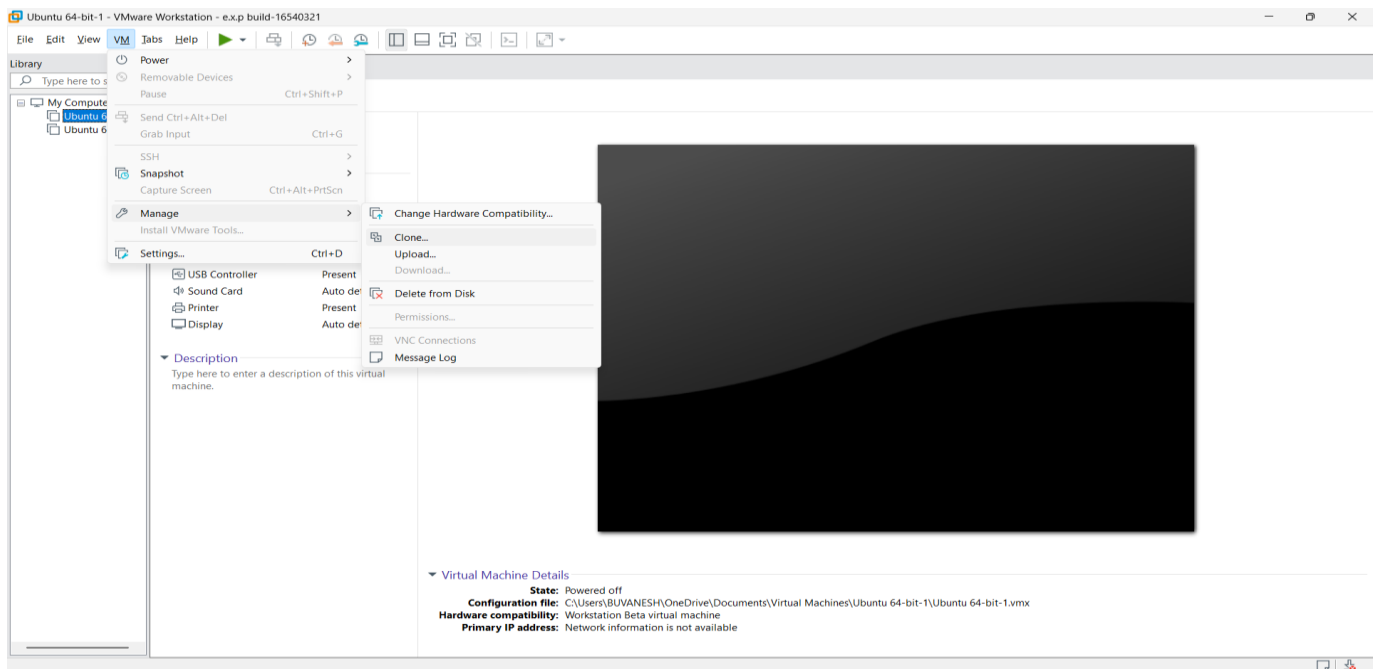
STEP 2: CREATE FILES ON DESKTOP.

STEP 3: CLICK ON VM AND SELECTS SNAPSHOT-> TAKE SNAPSHOT.

STEP 4: SNAPSHOT IS BEING DONE

IMPLEMENTATION:

STEP 1: GO TO VM AND GOTO MANAGE AND CLICK CLONE



STEP 2: CLICK CLONE

